

CVMAX: Accelerator Architecture with Polar Form Multiplication for Complex-Valued Neural Networks

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Abstract—Complex-Valued Neural Networks (CVNNs) have demonstrated high performance in applications where complex numbers are essential, but suffer from higher computational and memory overheads. Since their target applications often operate in resource-constrained environments, optimizing CVNNs for energy and area efficiency is important for their acceleration. To resolve these challenges, we present CVMAX, a software-hardware co-design for energy and area-efficient CVNN acceleration. CVMAX introduces a specialized quantization technique based on polar form representation and shift quantization. The technique significantly reduces the bit width of CVNNs and computational complexity compared to conventional quantization with rectangular form. Moreover, shift quantization leverages the computational simplicity of multiplication in polar form, reducing the complexity of complex number multiplication. With the quantization technique, we designed a dedicated hardware accelerator that supports CVMAX data and its associated arithmetic operations. In our evaluation, CVMAX achieves a 75% reduction in energy consumption and achieves a $4.44\times$ speedup compared to conventional accelerators.

1. Introduction

Complex Valued Neural Networks (CVNNs) are designed to effectively interact with applications that require complex numbers in their data. By leveraging the unique properties of complex numbers, CVNNs outperform conventional Real Valued Neural Networks (RVNNs) in tasks such as signal processing, MRI, and RADAR [1], [2], [3], [4], [5]. However, this advantage incurs significant computational and memory overhead. The arithmetic of complex numbers is inherently more computationally intensive, posing challenges for CVNNs [6]. Furthermore, target applications of CVNNs often run in resource-constrained environments, such as mobile devices, which require high energy and area efficiency to accelerate CVNNs in the hardware.

To address these challenges, we focus on polar form representation of complex numbers, a mathematical format for complex numbers. In polar form, a complex number is represented as $r \times e^{i\theta}$, where r denotes the magnitude and θ represents the phasor. Compared to the rectangular form ($a + bi$), polar form offers two distinct characteristics. First, magnitude and phasor have constrained value ranges; magnitude is always positive and the phasor falls within the range of 0 to 2π . These limited value ranges enable

more efficient quantization, reducing errors and improving compression rates [6]. However, prior work has only applied polar form quantization to memory optimization, without leveraging it for computation.

Second, a multiplication in polar form involves a simpler computation than in the rectangular form. In rectangular form, multiplication between two complex numbers requires four multiply, one add, and one subtract operation. In contrast, polar form only needs one multiply and one add operation to complete complex number multiplication. Despite the lower computational overhead of polar form in multiplication, polar form exhibits a disadvantage in addition. For the addition process, polar form necessitates conversion to the rectangular form. As a result, polar form and rectangular form have their own trade-off in their arithmetic.

Building on these characteristics, we propose CVMAX: an efficient accelerator architecture for Complex-Valued Neural Networks, incorporating polar form multiplication and shift quantization. CVMAX introduces the special co-design of the quantization and the hardware for efficient conversion between rectangular and polar forms. In CVMAX, we employ specialized shift quantization for the magnitude data of polar form, eliminating the need for the multiply operation in conversion process and complex number arithmetic. While conventional shift quantization has drawbacks in complex addition operations, we find that applying shift quantization solely to magnitude data avoids these issues and enables replacing multiplication with shift or add operations in complex number multiplication and conversion. With the quantization, we also design an accelerator architecture that performs complex number multiplication only with shifter and adder logic then utilize adder tree for complex number addition.

We summarize the contributions of CVMAX as follows:

- CVMAX leverages polar form quantization, capitalizing on its advantageous value range. For efficient complex number arithmetic, we adopt shift quantization for the magnitude part of the data.
- The CVMAX accelerator exploits simple polar form multiplication and a rectangular form adder tree in its processing element (PE) design. With CVMAX quantization, the PE of the CVMAX accelerator replaces all multipliers with shifters and adders, reducing both area and energy consumption.
- With CVMAX quantization, the CVMAX accelerator reduces PE area by 74% compared to conventional RVNN

Methods	Bit-width	Accuracy	Error
Baseline (Rectangular Form)	A 16/16 W 16/16	92.26%	-
Rectangular Form Quantization	A 8/8 W 8/8	90.98%	1.28%
Rectangular Form Quantization	A 8/8 W 4/4	90.40%	1.86%
Rectangular Form Quantization	A 4/4 W 4/4	84.28%	7.98%
Polar Form Quantization	A 8/8 W 8/8	91.44%	0.82%
Polar Form Quantization	A 8/8 W 4/4	90.81%	1.45%
Polar Form Quantization	A 4/4 W 4/4	88.12%	4.14%

Figure 1: Inference results of a CVNN model [13] applying uniform quantization in rectangular and polar forms at various bit widths. A refers to activation and W refers to weight. As complex number data consists of two data, we denote bit width of each data in activation and weight.

accelerators. In an iso-area comparison, the CVMAX accelerator achieves a $4.44\times$ speedup and 75% lower energy consumption compared to other accelerators.

2. Background and Motivation

A CVNN is a neural network that processes data as complex numbers and applies complex arithmetic in its layer operations [7], [8]. In certain applications, including MRI, RADAR, and sensor data processing, data is inherently represented as complex numbers due to theoretical foundations in those domains [9], [10]. In these areas, conventional RVNNs that operate only with real numbers would lose the information in the imaginary component and result in higher losses, although they also handle imaginary data as real numbers [10], [11], [12]. To address this, CVNNs are proposed to handle both the real and imaginary components by using complex numbers for their parameters and applying complex arithmetic in their layer operations and activation functions. As a result, CVNNs achieve higher performance in applications requiring complex-valued data [12]. However, CVNNs incur significant memory and computational overhead compared to RVNNs due to the increased complexity of handling complex-valued parameters and performing complex arithmetic [6].

Complex numbers are typically represented in two formats: rectangular form and polar form. In the rectangular form, a complex number is expressed as $a + bi$, where a and b are the real and imaginary components, respectively. In contrast, polar form represents a complex number using its magnitude and phase, expressed as $r \times e^{i\theta}$, where r is the magnitude and θ is the phasor. In the rectangular form, it does not have value limitation, so real and imaginary number can range from negative infinity to positive infinity. Unlike the rectangular form, polar form constrains values such that the magnitude is always positive, and the phasor ranges from 0 to 2π radian. Although both forms represent the same data, the narrower value range of polar form provides certain advantages, particularly in quantization.

Previous work [6] demonstrated that the narrow value range of polar form makes it more advantageous for quantization compared to rectangular form. The performance of quantization depends heavily on factors such as bit width, value range, and value distribution. Due to its narrower value range, polar form can reduce the required bit width or

Matrix Multiplication

$$c_{real} = \sum_{n=1}^n (a_{real} \times b_{real} - a_{img} \times b_{img}) \quad \text{Complex Number Multiplication} \rightarrow 4 \text{ multiply} + 2 \text{ add / subtract}$$

$$c_{img} = \sum_{n=1}^n (a_{real} \times b_{img} + a_{img} \times b_{real}) \quad \text{Complex Number Addition} \rightarrow 2 \text{ add}$$

$$\text{Total} = N \times \text{Complex Number Multiplication} + (N-1) \times \text{Complex Number Addition} = N \times (4 \text{ multiply} + 2 \text{ add/subtract}) + (N-1) \times 2 \text{ add}$$

(a) Rectangular Form

Matrix Multiplication

$$c_m e^{ic_\theta} = \sum_{n=1}^n a_m \times b_m e^{i(a_\theta + b_\theta)} \quad \text{Complex Number Multiplication} \rightarrow 1 \text{ multiply} + 1 \text{ add}$$

$$= \sum_{n=1}^n d_m e^{id_\theta} \quad \text{Complex Number Conversion} \rightarrow 2 \text{ multiply} + 2 \text{ sine/cosine}$$

Conversion

$$d_{real} = d_m \times \cos(d_\theta) \quad \text{Complex Number Addition} \rightarrow 2 \text{ add}$$

$$d_{img} = d_m \times \sin(d_\theta)$$

$$\text{Total} = N \times (\text{Complex Number Multiplication} + \text{Complex Number Conversion}) + (N-1) \times \text{Complex Number Addition} = N \times (3 \text{ multiply} + 1 \text{ add} + 2 \text{ sine/cosine}) + (N-1) \times 2 \text{ add}$$

(b) Polar Form

Figure 2: Arithmetic operations for computing a single output element in complex matrix multiplication using (a) rectangular form and (b) polar form. For simplicity, we derive the operations only for a single output element.

achieve lower quantization loss than rectangular form. However, prior work only applied quantization to store weights in memory and performed layer operations using decompressed weights and activations. To evaluate the potential of extending quantization to actual layer operations, we conducted an initial analysis of accuracy, as shown in Fig. 1. In this analysis, we applied basic symmetric quantization to both activations and weights of a CVNN model [13] in rectangular and polar forms at three different bit widths. The results show that quantization in polar form consistently achieves higher accuracy across all bit widths and retains accuracy at lower bit widths compared to rectangular form. With the analysis, we can verify that quantization with polar form is effective not only for weights but also for activations.

Since complex number matrix multiplication is the primary computational bottleneck in CVNNs, we also analyzed how arithmetic operations differ between the two formats. Fig. 2a shows the total arithmetic operations of two representation formats for a single output element of a complex number matrix multiplication. As depicted in Fig. 2a, the rectangular form requires two add operations for its add operation between complex numbers (e.g., $(a+bi) + (c+di) = (a+c) + (b+d)i$). However, its multiply operation requires four multiply, one add, and one subtract. In consequence, a single output element of N by N requires total $N \times (4 \text{ multiply} + 2 \text{ add/subtract}) + (N-1) \times 2 \text{ add}$.

In contrast, polar form exhibits different characteristics in their arithmetic. Unlike the rectangular form, an addition in polar form involves a more complex process that requires a conversion to the rectangular form. Fig. 2b shows that an addition in polar form involves two trigonometric operations and two multiply operations for conversion. After

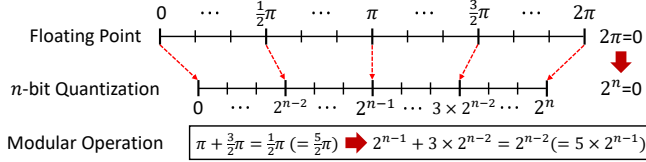


Figure 3: CVMAX quantization for phasor number with modular characteristics.

conversion, polar form addition proceeds in the same way as in rectangular form. To avoid the conversion, the previous work [6] adopts rectangular form for matrix multiplication, even though it uses polar form for the quantization.

However, we find that polar form is advantageous for multiplication. Polar form multiplication operations require only one multiply and one add operation. With this simpler multiplication process, polar form require $N \times (3 \text{ multiply} + 1 \text{ add} + 1 \text{ sine} + 1 \text{ cosine}) + (N - 1) \times 2 \text{ add}$ for the same matrix multiplication. Considering that trigonometric operations can be simplified in hardware as in previous work [6], polar form consumes fewer multiply operation than rectangular form, even with the conversion.

3. CVMAX

With the analysis, we propose CVMAX: an efficient accelerator architecture with polar form quantization and multiplication for Complex-Valued Neural Networks. CVMAX targets polar-form matrix multiplication and introduces a co-designed specialized quantization technique and accelerator architecture. To leverage the simpler multiplication inherent to polar form, the CVMAX accelerator executes complex number multiplications using quantized polar-form data. We design the accelerator with an adder tree architecture, which can concurrently converts the data into rectangular form and sums the results. For a fast and lightweight matrix multiplication, CVMAX introduces specialized shift quantization, which applies shift quantization to the magnitude data of activations and weights. This approach eliminates all multiplication operations, replacing them with shift operations. Consequently, these techniques enable the CVMAX accelerator to replace multipliers with shifters and adders in its PE design, leading to significant reductions in both area and energy consumption. Moreover, the CVMAX quantization method, which leverages the benefits of polar form to reduce quantization error and loss, achieves superior performance compared to rectangular-form quantization. In the following subsections, we provide a detailed description of the CVMAX quantization method and the architectural design of the CVMAX accelerator.

3.1. CVMAX Quantization

To exploit the advantages of polar form in quantization, CVMAX quantization applies specialized quantization to the magnitude and phasor data. For phasor data, we extend the quantization method from previous work [6] by leveraging the modular characteristics of the phasor, as shown in Fig. 3. Unlike conventional uniform symmetric quantization, we define the value range of phasor as 0 to 2π with $2^{\text{bitwidth}} + 1$

Matrix Multiplication

$$\begin{aligned} 2^c e^{i c \theta} &= \sum_{m=0}^n 2^{a_m} \times 2^{b_m} e^{i(a_m + b_m \theta)} \\ &= \sum_{m=0}^n 2^{(a_m + b_m)} e^{i(a_m + b_m \theta)} \\ &= \sum_{m=0}^n 2^{d_m} e^{i d_m \theta} \end{aligned}$$

Conversion

$$\begin{aligned} d_{\text{real}} &= 2^{d_m} \text{array}_{\cos}[d_\theta] \\ &= \text{array}_{\cos}[d_\theta] \gg d_m \\ d_{\text{img}} &= 2^{d_m} \text{array}_{\sin}[d_\theta] \\ &= \text{array}_{\sin}[d_\theta] \gg d_m \end{aligned}$$

Complex Number Multiplication
→ 1 multiply + 1 add
→ 2 add (CVMAX)

Complex Number Conversion
→ 2 multiply + 2 sine/cosine
→ 2 shift + 1 lookup table (CVMAX)

Complex Number Addition
→ 2 add

Total

$$\begin{aligned} &N \times (\text{Complex Number Multiplication} + \text{Complex Number Conversion}) \\ &+ (N - 1) \times \text{Complex Number Addition} \\ &= N \times (2 \text{ add} + 1 \text{ lookup table} + 2 \text{ shift}) + (N - 1) \times 2 \text{ add} \end{aligned}$$

Figure 4: CVMAX matrix multiplication.

bins rather than 2^{bitwidth} . This modification accounts for the circular nature of the phasor, where 2π is equivalent to 0. By using $2^{\text{bitwidth}} + 1$ bins, CVMAX can effectively represent special values within the phasor, as illustrated in the figure. Since the outputs of trigonometric functions are zero for these special values, representing those numbers in quantization is critical for accuracy. While conventional quantization with 2^{bitwidth} bins fails to capture all these special values with uniform quantization, CVMAX quantization can reduce the bit width required for phasor data while preserving accuracy by leveraging this modular design.

In complex number matrix multiplication, the magnitude involves multiply operations between magnitudes and trigonometric values, as shown in Fig. 2b. To eliminate these multiply operations, we focus on shift quantization [14], which replaces multiplications with shift operations. In RVNNs or rectangular form data, applying shift quantization to both weight and activation often leads to high accuracy loss and complicated addition between shift-quantized data. However, we focus on that magnitude numbers do not necessitate addition operations with each other during complex number matrix multiplication. With this characteristic, we find out that applying shift quantization exclusively to the magnitude of both weights and activations avoids these challenges. Moreover, since the magnitude is always a positive number, shift quantization in CVMAX does not require a sign bit for negative values, resulting in further bit-width savings. With the observation, CVMAX employs shift quantization for the magnitude of both weights and activations, effectively replacing multiply operations between magnitudes with add operations and multiply operation with the results of trigonometric functions with shift operations, as shown in Fig. 4.

With this specialized quantization, CVMAX changes the arithmetic operations required for complex matrix multiplication, as illustrated in Fig. 4. For each complex number multiplication, CVMAX requires two add operations: one for adding the phasors and another for combining the magnitudes. Since magnitudes in CVMAX are represented as exponents due to shift quantization, the multiplication

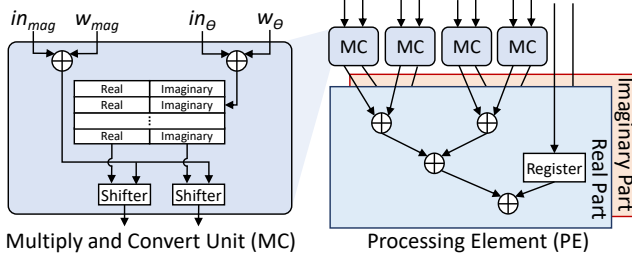


Figure 5: Single PE design of the CVMAX accelerator. The PE comprises a Multiply and Convert (MC) unit and a coupled adder tree unit.

of two magnitudes is transformed into an addition of their exponents. The conversion process involves a lookup table search and two shift operations. CVMAX adopts the lookup table optimization from previous work [6] for trigonometric operations, enabling the use of a single dual-output lookup table. This approach replaces complex operations with one lookup table search and minimizes computation overhead. Multiplications between the results of trigonometric functions and magnitudes are also converted into shift operations using CVMAX quantization. As a result, CVMAX involves total $N \times (2 \text{ add} + 1 \text{ lookup table search} + 2 \text{ shift}) + (N - 1) \times 2 \text{ add}$ for a single output element of matrix multiplication. Since multiplications are computationally expensive than additions or shift operations, CVMAX effectively reduces computational complexity of CVNN layers.

In addition, CVMAX also designs the quantization for rectangular form, as complex number addition is performed in rectangular form. In CVMAX, the rectangular form is used for representing the result of trigonometric functions, ranging from -1 to 1, and multiplication results between the results and magnitude. Consequently, the value ranges of these results are exactly twice as large as the value range of magnitudes. CVMAX offers the flexibility to assign different bit widths to rectangular-form data compared to phasor data by configuring the bit width of the stored results of trigonometric functions. In our default configuration, the bit width of the stored results is set at 8-bit, while the bit widths for both phasor and magnitude data are set to 4 bits. Using a 4-bit magnitude allows up to 8 bits of shift operations to be applied to an 8-bit result of trigonometric function, resulting in 16-bit rectangular-form data for the addition process. By allocating more bit width to the results of trigonometric functions and addition operations, CVMAX reduces the size of polar-form data, which is critical for minimizing the area and energy consumption of the accelerator. This design strategy enables CVMAX quantization to maintain accuracy comparable to 8-bit quantization in other methods, while employing only 4-bit magnitudes and 4-bit phasors.

3.2. CVMAX Accelerator

The CVMAX accelerator hardware is specifically designed to take full advantage of the optimizations introduced by CVMAX quantization, particularly through the use of shift operations and the elimination of multipliers. The accelerator consists of adder tree-based PEs and activation units.

The CVMAX accelerator executes CVNN layer operations using the specialized PEs, while activation functions are handled by the activation unit. The PE features a specialized adder tree architecture that follows the CVMAX matrix multiplication process.

Fig. 5 illustrates the detailed design of a single PE in the CVMAX accelerator. The PE is implemented as an adder-tree-based multiply and accumulate (MAC) unit, consisting of two key components: the Multiply and Convert (multiply and convert (MC)) unit and the adder tree. The MC unit performs complex number multiplication using CVMAX data by applying polar-form multiplication and then converting the data into rectangular form. Since CVMAX quantization replaces multiplication between magnitudes with addition, the MC is composed of two adders, two shifters, and one dual-output lookup table, eliminating the need for multipliers. For complex number multiplication, the MC unit adds the magnitudes and phasors from the inputs and weights, as shown in Fig. 4. Using the added phasor, the MC probes precomputed value of trigonometric functions in the lookup table and retrieves two outputs corresponding to the real and imaginary components. Finally, the MC applies shift operations to both data using the added magnitude. Through these operations, the MC performs complex number multiplication in polar form and converts the data into rectangular form.

The adder tree within the PE handles the complex number additions in rectangular form using the outputs generated by the MCs. To sum the real and imaginary components of the complex numbers, the adder tree is designed as a coupled structure: one part processes the real components, while the other processes the imaginary components. Each adder tree receives its respective inputs from the MCs and computes the partial sums for each component. This coupled design allows the adder tree to produce two outputs, representing the partial sums of the real and imaginary components.

In CVMAX quantization, the default bit width for both the magnitude and phasor is 4 bits. As a result, the CVMAX accelerator employs 4-bit adders in the MC unit along with a 16-entry lookup table. To accommodate with the 8-bit trigonometric value, the lookup table is configured with an 8-bit width, while the adder tree operates with corresponding bit widths for its outputs. This configuration optimizes the hardware utilization by allocating higher bit widths to certain logic while keeping the phasor and magnitude at lower bit widths. This design minimizes hardware overhead while preserving the accuracy of CVMAX. By replacing traditional multipliers with lookup tables, adders, and shifters in the MC unit, the CVMAX accelerator achieves single-cycle operation for the PE. This reduces the number of cycles required for matrix multiplication compared to conventional rectangular-form implementations.

While the PE unit accelerates complex number matrix multiplications in CVNN layers, the activation unit handles other operations required by CVNNs. First, the activation unit computes activation functions specific to CVNNs (e.g., modRelu, zRelu, and cardioid). Additionally, it performs conversions between polar and rectangular forms. This conversion process is integrated with requantization, a critical

TABLE 1: CVNN Benchmarks for Quantization

Model	Tasks	Dataset
CVNN_1 [18]	Image Generation	MRI Data [19]
CVNN_2 [13]	PolSAR classification	PolSF [20]

step for accelerators that use quantization. By combining the conversion and requantization processes, the CVMAX activation unit eliminates the need for separate operations, streamlining overall execution.

The CVMAX accelerator dynamically determines the sequence of conversion and activation function execution based on the required format by the activation function. For instance, if the activation function requires polar-form data, the activation unit first performs the conversion process before applying the activation function. Conversely, if the activation function operates in rectangular form, the unit executes the activation function first and then converts the data into polar form for subsequent layer operations. Since these two operations take fewer cycles than matrix multiplication executed on the PE units, it operates seamlessly within the pipelined architecture, avoiding stall cycles.

4. Evaluation

4.1. Methodology

To evaluate the quantization of CVMAX, we implemented CVMAX quantization in two CVNN models using the PyTorch [15] and TensorFlow [16] frameworks. Table 1 lists the CVNN models and their corresponding datasets used in our evaluation. Each model was evaluated on a specific task, with performance metrics tailored to the nature of the task. For CVNN_1 [17], an image generation task, the performance metric used is Structural Similarity Index Measure (SSIM), where higher values indicate better performance. In contrast, for CVNN_2 [13], a classification task, we used accuracy. We adopted the dataset and evaluation metric from original papers [13], [17].

We conducted a comparative evaluation of the CVMAX accelerator against other RVNN accelerators using a cycle-accurate simulator. To compare with various types of accelerators, we evaluated two RVNN accelerator designs: a systolic-array-based accelerator and an adder-tree-based accelerator. For the systolic-array-based accelerator, we selected a TPU-like [21] model based on an open-source simulator [22]. For the adder-tree-based accelerator, we used a tensor core-like model based on Accel-Sim [23]. Additionally, we evaluated the accelerator from previous work [6], which is a systolic-array-based design targeting CVNNs. For executing CVNN on RVNN accelerator, we assume that each CVNN layer is split to four sub-layer as described in previous work [6]. To measure the area and energy consumption of the PE and activation unit, we used the Synopsys Design Compiler with the FreePDK 45nm library and CACTI [24].

4.2. Evaluation on Quantization

To assess the performance of quantization, we set the baseline models to use FP16 for their real and imaginary

Quantization Methods	Bit-width	CVNN_1		CVNN_2	
		SSIM	Error	Accuracy	Error
Baseline	A16/16 W16/16	0.910	-	92.26%	-
Rectangular From	A8/8 W8/8	0.891	0.019	90.98%	1.28%
Rectangular From	A4/4 W4/4	0.821	0.089	84.28%	7.98%
Polar From	A8/8 W8/8	0.897	0.013	91.44%	0.82%
Polar From	A4/4 W4/4	0.855	0.055	88.12%	4.14%
PAQ	A8/8 W8/8	0.899	0.011	91.51%	0.75%
PAQ	A4/4 W4/4	0.860	0.050	89.99%	2.27%
CVMAX	A8/8 W8/8	0.898	0.012	91.49%	0.77%
CVMAX	A4/8 W4/8	0.890	0.020	91.01%	1.25%
CVMAX	A4/4 W4/4	0.871	0.039	90.00%	2.26%

Figure 6: Inference results of CVNN models [13], [18] applying various quantization methods across various bit widths. A refers to activation and W refers to weight.

data and evaluated them using four quantization methods. The first two quantization methods are basic quantization methods applied in rectangular and polar form, as described in Section 2. The remaining two are specialized quantization methods designed specifically for CVNNs. The first of these is PAQ [6], a prior work that applies quantization in polar form. While the original implementation only applies quantization to weight data, we extended it to include both activations and weights for a fair comparison with other methods. The final method is the proposed CVMAX quantization. In our evaluation, we considered two bit-width configurations for the quantization methods. We applied 8-bit and 4-bit for each data to the quantization methods, with 16-bit and 8-bit used for addition operations, respectively. For CVMAX quantization, we introduced an additional bit-width configuration: a 4-bit magnitude combined with an 8-bit phasor. This configuration was chosen because a 4-bit shift-quantized magnitude in CVMAX can represent shifts ranging from 8-bit shift right to an 8-bit shift left, effectively generating 16-bit data for the addition process.

Fig. 6 provides the inference results of our benchmarks with the above quantization methods. As shown in the figure, the CVMAX quantization design with 4-bit for each data outperforms the other quantization methods using 4-bit bit-width. These results stem from the characteristics of shift quantization, which benefits from covering a larger value range with fewer bits. As a result, the CVMAX quantization design maintains accuracy within a 1% error margin compared to 8-bit rectangular form quantization. Besides, as the shift quantization exhibit poor accuracy preserving in larger bit widths, CVMAX shows lower accuracy than PAQ in the design with 8-bit for each data. Furthermore, CVMAX quantization with 4-bit magnitude and 8-bit phasor achieves comparable performance with other quantization of 8-bit, while reducing data size by 25% through the reduction of the magnitude to 4-bit. These results highlight the effectiveness of CVMAX quantization in reducing bit width while maintaining an acceptable error rate.

4.3. Area Analysis

We compared the area of the CVMAX accelerator with conventional accelerators. For a fair comparison, we standardized the computational capabilities of the accelerators to a 64 by 64 systolic array in a TPU-like architecture.

TABLE 2: Area Comparison

	Bit width	Relative Area
TPU-like	A8/8 W8/8	1.00
Tensor Core-like	A8/8 W8/8	0.78
PAQ Accelerator	A8/8 W8/8	1.02
CVMAX Accelerator	A4/4 W4/4	0.26

TABLE 3: Hardware configuration for iso-area comparison

	# PEs	# Multiplications
TPU-like	4096	4096
Tensor Core-like	1310	5240
PAQ Accelerator	4015	4015
CVMAX Accelerator	962	20960

Since one PE of the CVMAX accelerator can execute four complex number multiplications—which would require 16 multiplications in RVNN accelerators—we set the number of CVMAX PEs to 256. Similarly, we set the number of tensor cores to 1024, as each core executes four real number multiplications. Table 2 shows the results of our area comparison. The CVMAX accelerator achieves a 74% and 65% area reduction compared to the TPU-like accelerator and the tensor core-like design and 75% area reduction compared to the PAQ accelerator.

4.4. Performance and Energy Analysis

To ensure a fair comparison for performance and energy, we conducted the evaluations based on iso-area criteria across four distinct accelerator designs. The area configuration for performance evaluation is detailed in Table 3. According to Fig. 6, we adjusted the bit widths of the four accelerators to maintain accuracy close to that achieved with 8-bit rectangular form quantization, the quantization method used in TPU-like and tensor core accelerators. For the iso-area comparison, we set the accelerators to have the same area as a 64 by 64 TPU-like systolic array. We set the number of CVMAX PEs to 962, which can execute 20960 real number multiplications at once, and configured the other accelerators similarly. Furthermore, to broaden the scope of evaluation, we included two additional benchmarks for performance analysis, referencing virtual CVNN models generated from conventional RVNN models. For the models, we convert all data of models into complex number and layers to its counter part layer in CVNN.

Fig. 7 illustrates the relative performance of CVNNs in various accelerator architectures. Notably, the CVMAX accelerator demonstrates a $4.44\times$ and $3.9\times$ speedup compared to the TPU-like model and the tensor core-like accelerator, and $3.1\times$ speedup compared to the PAQ accelerator, respectively. These performance gains come from the larger computational capabilities of the CVMAX accelerator which can execute about five times more multiplications than TPU-like models at once, as shown in Table 3.

We also evaluated the energy consumption of the accelerators, as shown in Fig. 8. The CVMAX accelerator reduces energy consumption by 75% and 73% compared to the TPU-like model and the tensor core-like accelerator, and 65% energy consumption than the PAQ accelerator, respectively. The energy savings of CVMAX are mainly achieved with

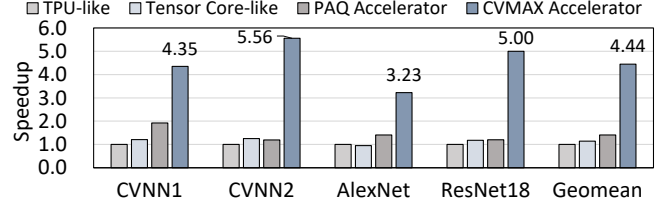


Figure 7: Relative performance of CVNNs in various accelerator architectures. All performance results are normalized to TPU-like model.

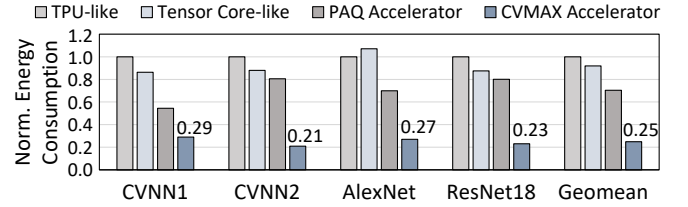


Figure 8: Relative energy consumption of CVNNs in various accelerator architectures.

reduced cycle times and reduced buffer and memory usage. In the evaluation, we adopted the same memory and buffer sizes for all accelerators. However, resizing the memory and buffer according to the reduced bit width could result in additional energy savings by lowering the static power of the memory and buffer and reducing the total area of the accelerator.

5. Conclusion

CVNNs exhibit high performance in applications that use complex numbers, but also incur significant computational and memory overheads due to their complex-valued data and layer operations. To mitigate these overheads, we propose CVMAX, a software-hardware co-design for energy- and area-efficient acceleration of CVNNs. CVMAX employs quantization techniques that effectively reduce bit widths and replace multiply operations with shift and add operations by adopting power-of-two-based shift quantization for magnitudes. This specialized quantization technique enables the CVMAX accelerator to support CVNN layer operations without a multiplier, which typically consumes high energy and occupies substantial area in the hardware budget. By eliminating the multiplier, CVMAX achieves a 75% reduction in energy consumption and a $4.44\times$ speedup in iso-area comparisons, while maintaining an acceptable error rate compared to conventional hardware and software designs.

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